2b. Content of A Level in Design and Technology (H404, H405 and H406)

Central to the content of this qualification is the requirement for learners to understand and apply processes of iterative designing in their design and technology practice. They will need to demonstrate their knowledge, understanding and skills through interrelated iterative processes that 'explore' needs, 'create' solutions and 'evaluate' how well the needs have been met.

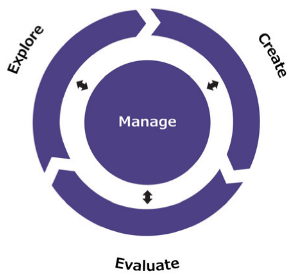

Fig. 1 Iterative Design Wheel

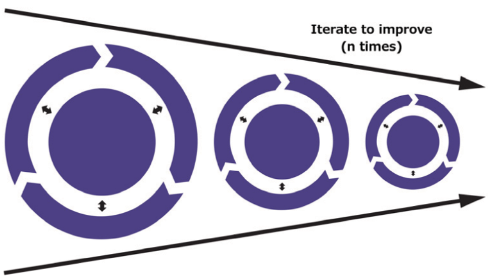

Fig. 2 Multiple iterations of design

At the centre of any iterative process is the need to develop critical-creative thinking skills to manage and organise opportunities that are identified. This learning will equip learners with life-long skills of problem spotting and problem solving, and enable them to apply their learning to different social, moral and commercial contexts.

The enquiry nature of this specification's content will encourage learners to make links between topics and to explore, create and evaluate a range of outcomes. It encourages a creative approach supported by subject knowledge in order to design and make prototypes that solve authentic, real-world problems and have real potential to become viable products.

The knowledge, understanding and skills that all learners must develop are underpinned by technical principles predominantly assessed in the written exam, and designing and making principles predominantly assessed in the non-exam assessment (NEA) although there is an expectation that learning builds a holistic understanding of the subject.

There is distinct content for the exam and non-exam assessment, but this is held together through nine topic areas that shape all components and give clarity, these are:

1. Identifying requirements
2. Learning from existing products and practice
3. Implications of wider issues
4. Design thinking and communication
5. Material considerations
6. Technical understanding
7. Manufacturing processes and techniques
8. Viability of design solutions
9. Health and safety.

Experiencing learning through practical activity, (both designing and technical principles) is fundamental to the delivery of this specification, as is the importance of the contextual relevance of design and technology practice. Learners should, as a result, be given increased autonomy to make decisions in order to justify their reasoning when solving problems in their own way.

The 'Iterative Design Project' is a substantial design, make and evaluate project that allows learners to reposition or develop further an existing product in relation to a given context. The experience of this will be supported by and support their learning for the 'Principles' written exam.

Design and technology requires learners to apply mathematical skills and understand related science. This reflects the importance of Design and Technology as a pivotal STEM subject. This specification along with prior learning in Design and Technology and other subjects offers the opportunity for learners to build on and apply their learning at Key Stage 4 and Key Stage 5.